Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SASW")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40, 2000);
Distrib (Fm,           TruncNormal,            0.1,      0.2,    0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,         0.8,      0.1,    0.1, 2);
Distrib (kap,          Uniform,                0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40, 300);
Distrib (E_coc,        LogUniform,              1,      300);
Distrib (rOM_ClxHorse,  LogUniform,           0.00001,      0.3);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcaliS DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	134.322000	0.44762300	1.07568000	0.6130560	119.85900	8.444190
2.5%	118.263000	0.34846600	0.98552745	0.5841350	94.47450	7.534500
97.5%	152.018000	0.61120800	1.15347750	0.6447690	148.88000	9.556736
Min	105.268000	0.08727340	0.81352300	0.2908240	75.44090	3.555110
Max	167.638000	0.73685700	1.19416000	0.8525060	164.05900	253.776000
Result_mean	134.142958	0.46796832	1.07146560	0.6136636	119.72465	8.516415
Result_sd	8.887887	0.06800257	0.04402798	0.0161127	14.25175	1.579918

```
             rOM_ClxHorse.1. Sigma_W.1.
Result_mode    0.0107414000 0.14551100
2.5%           0.0057518720 0.13304425
97.5%          0.0137956000 0.17078790
Min            0.0000106067 0.11807400
Max            0.1516470000 4.24484000
Result_mean    0.0103285547 0.15064238
Result_sd      0.0023261261 0.03653092

Number of observations, n =  277
Number of parameters, k =  7
Log-likelihood, LL =  -52.52611
BIC =  144.4203


Potential scale reduction factors:

                Point est. Upper C.I.
E_coc.1.              1.08       1.27
Ehp.1.                1.25       1.70
Fm.1.                 1.10       1.31
kap.1.                1.20       1.58
pAm.1.                1.23       1.63
r_pAm_pM.1.           1.21       1.59
rOM_ClxHorse.1.       1.02       1.03
Sigma_W.1.            1.03       1.11

Multivariate psrf

1.18
```

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

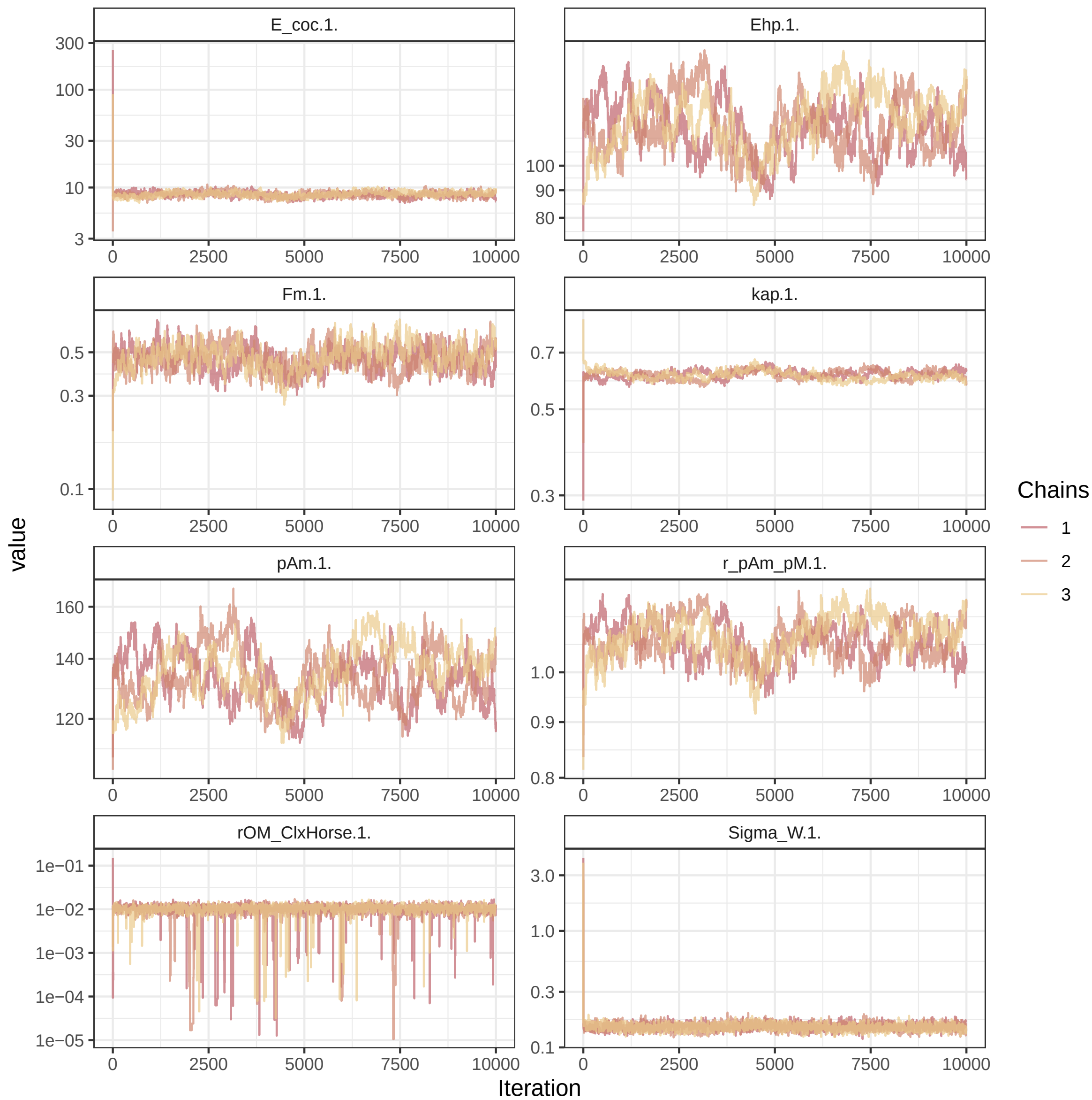


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

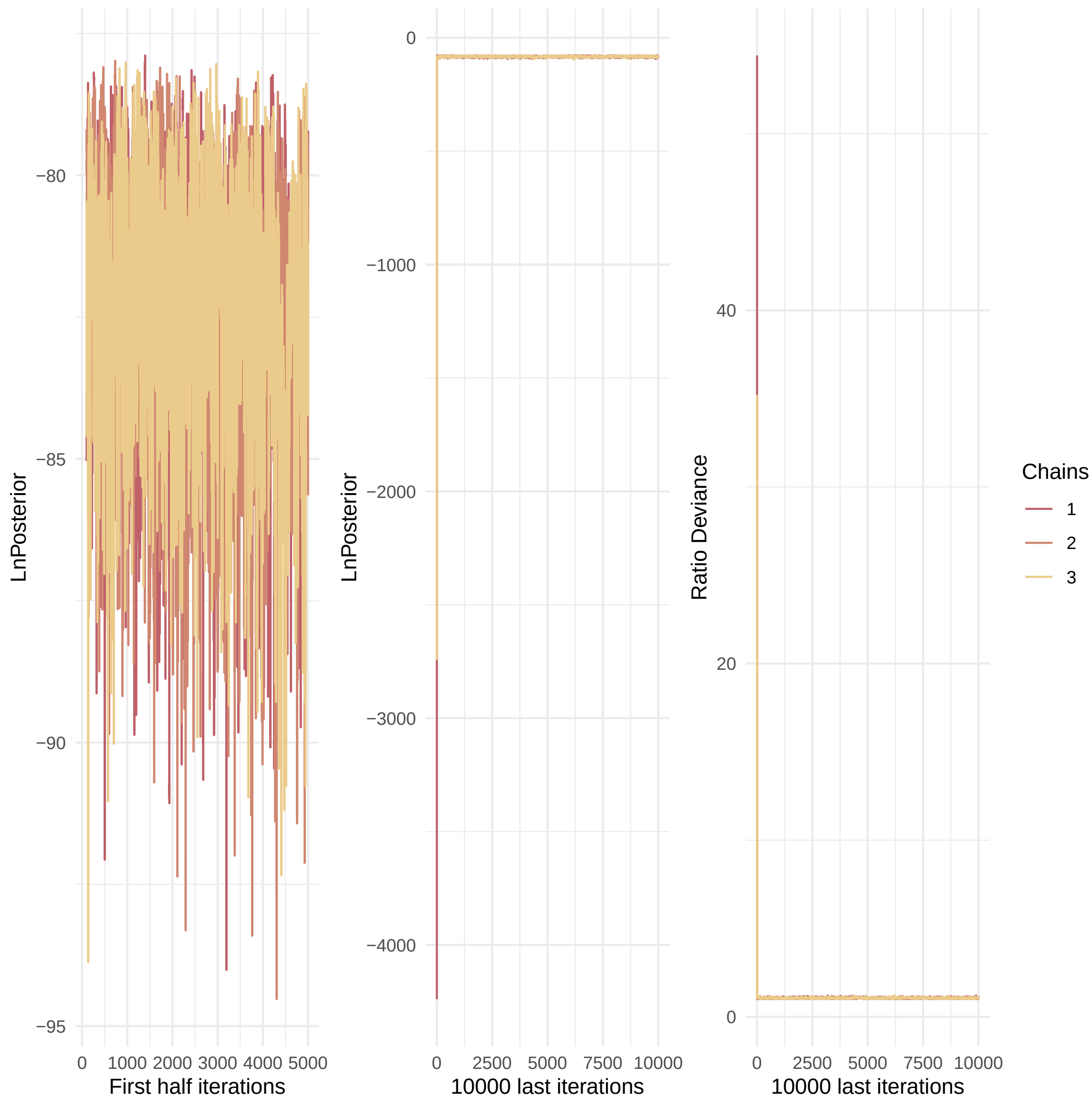


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

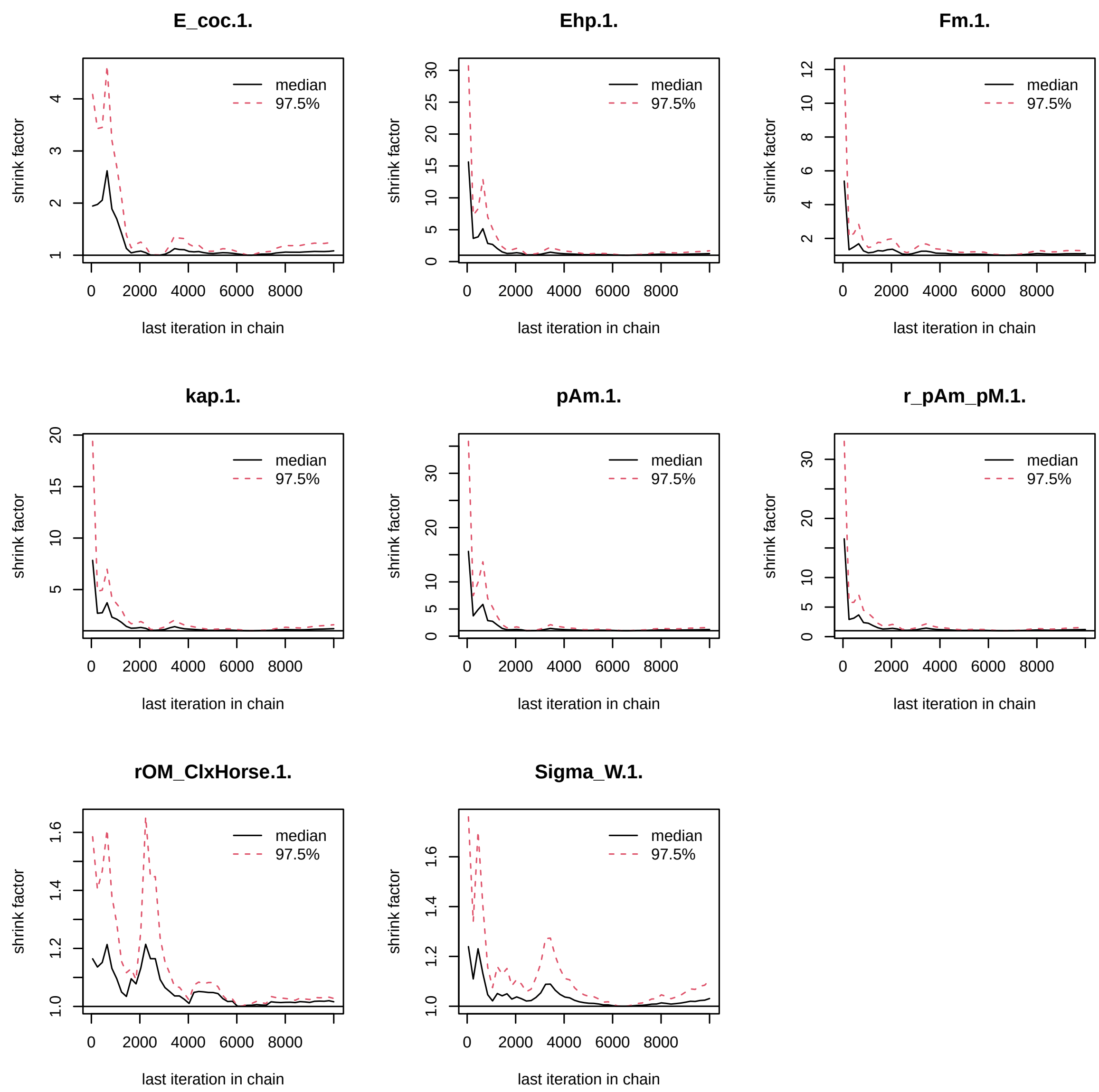


Figure 3: Chains convergence.

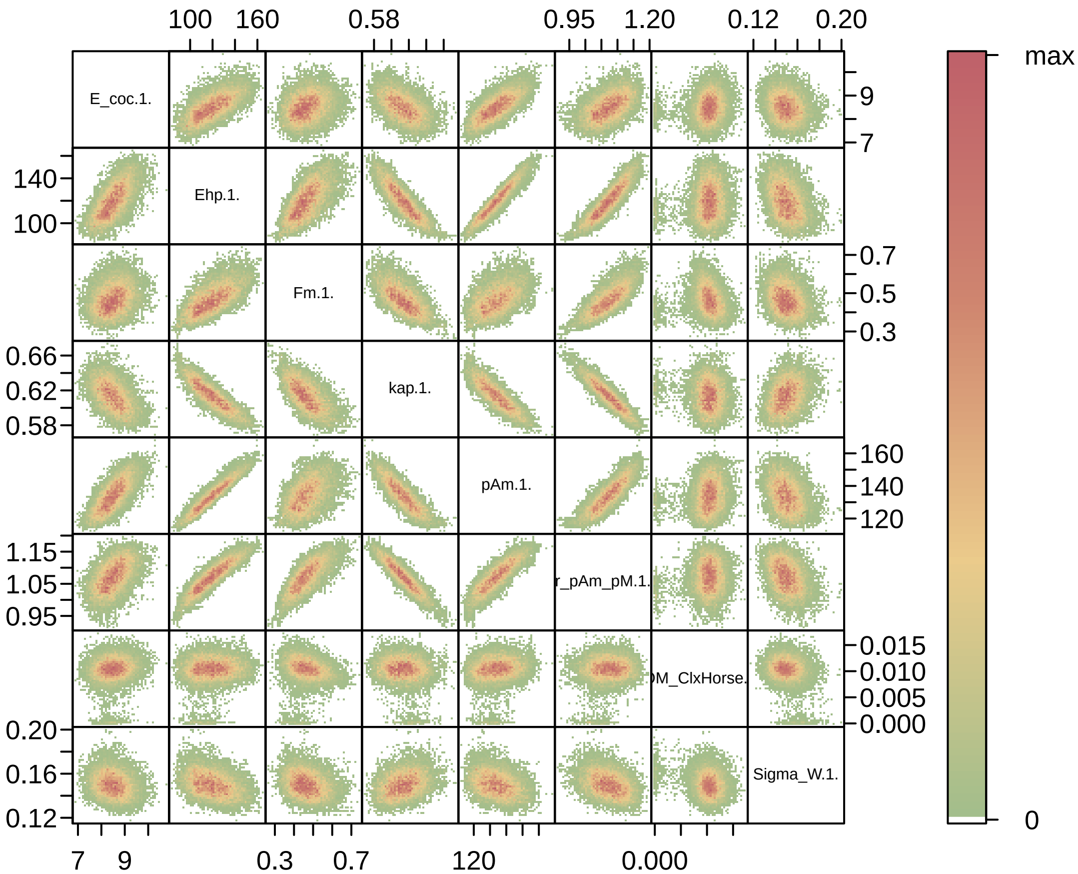


Figure 4: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

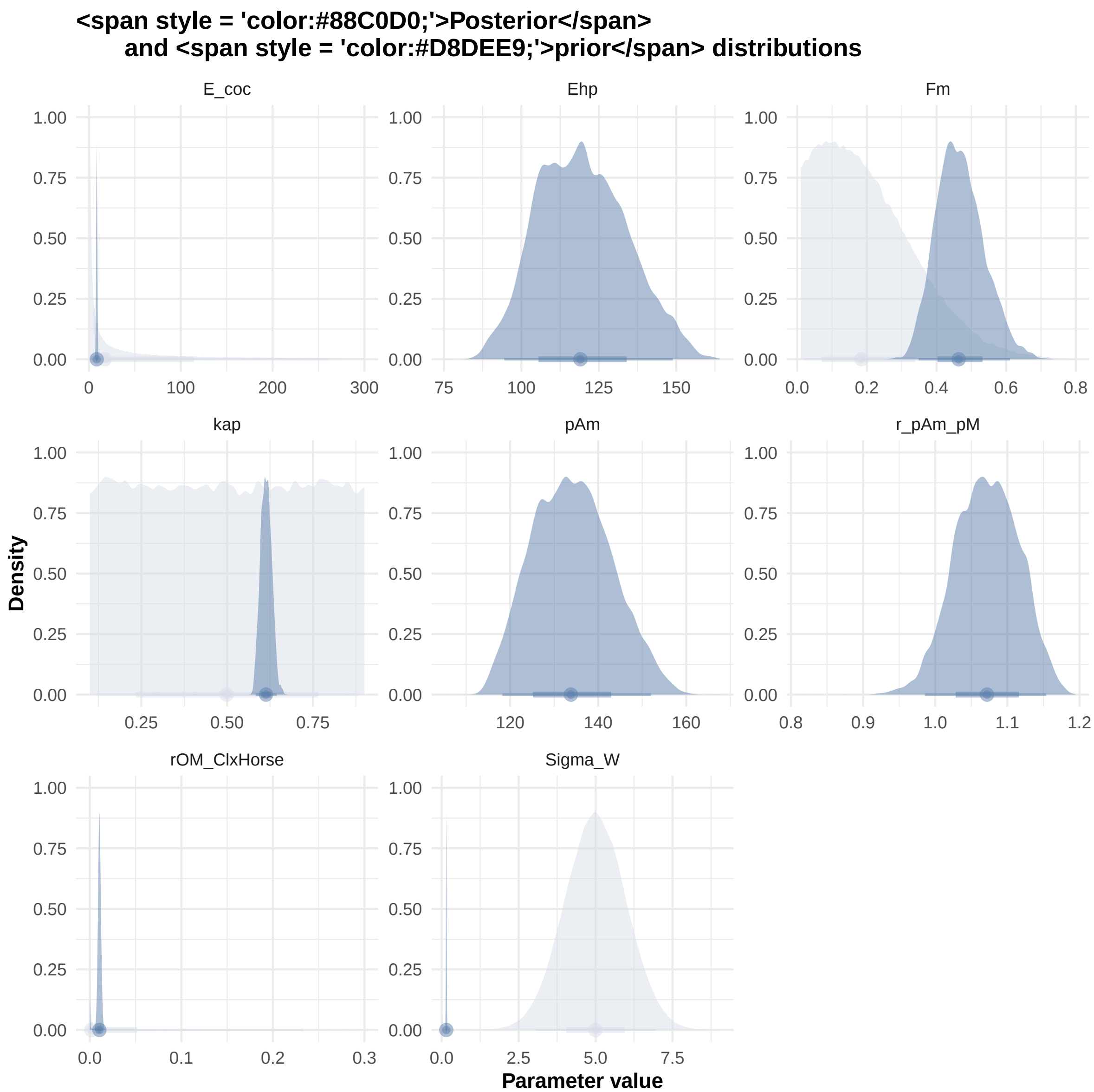


Figure 5: Distributions of the priors and the posteriors.

```r
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Reproduction"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcaliSL DEB_sim_full.in
```

```r
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
```

```r
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.
```

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

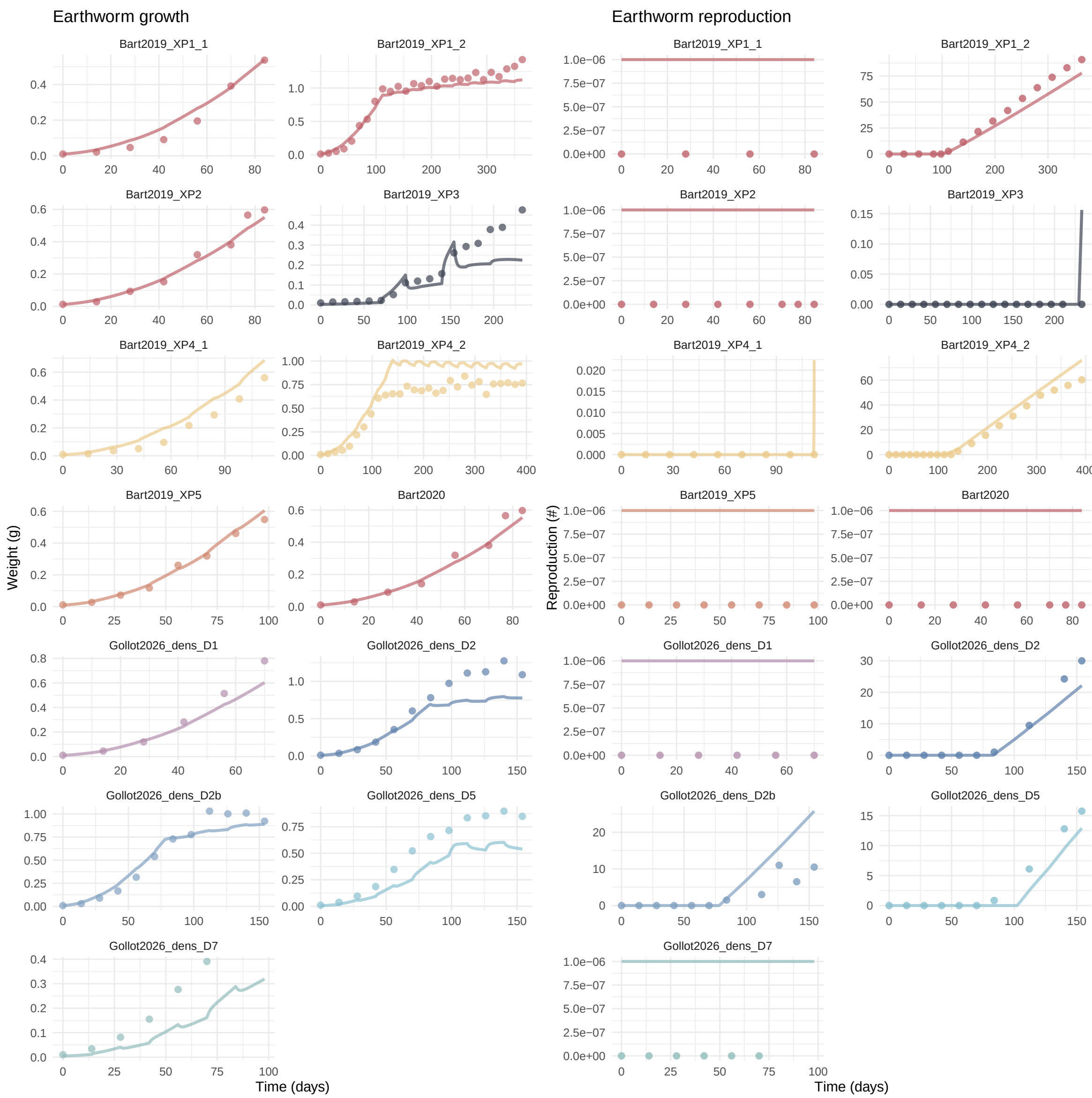
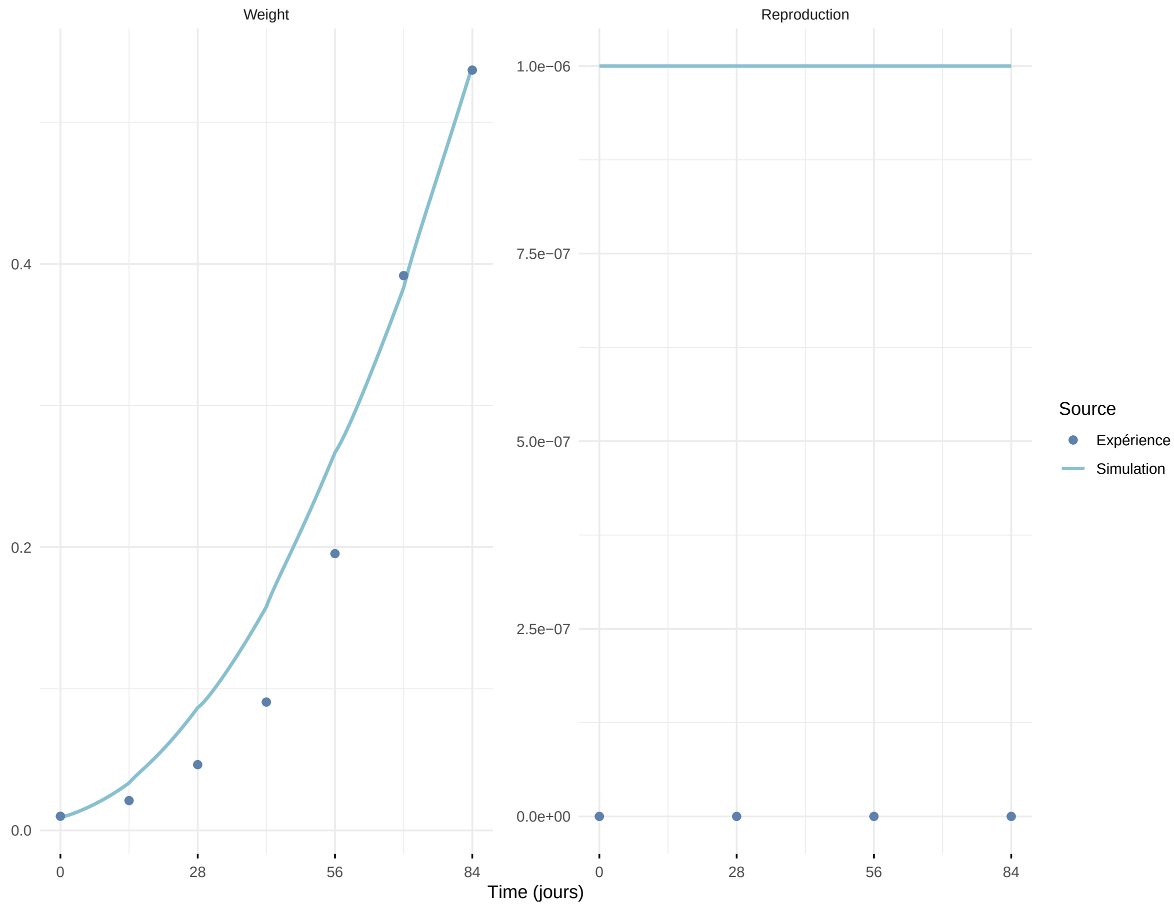
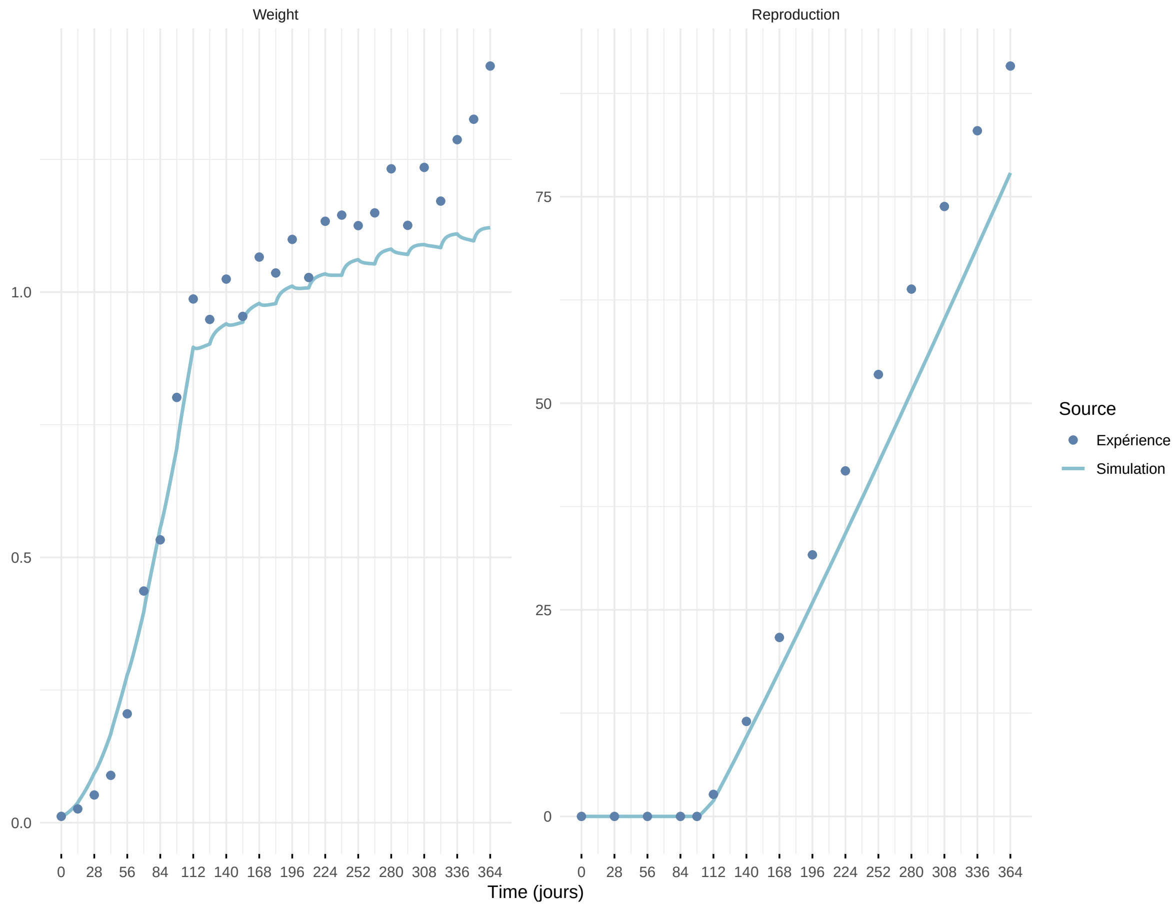


Figure 6: Simulations Weight and Reproduction.

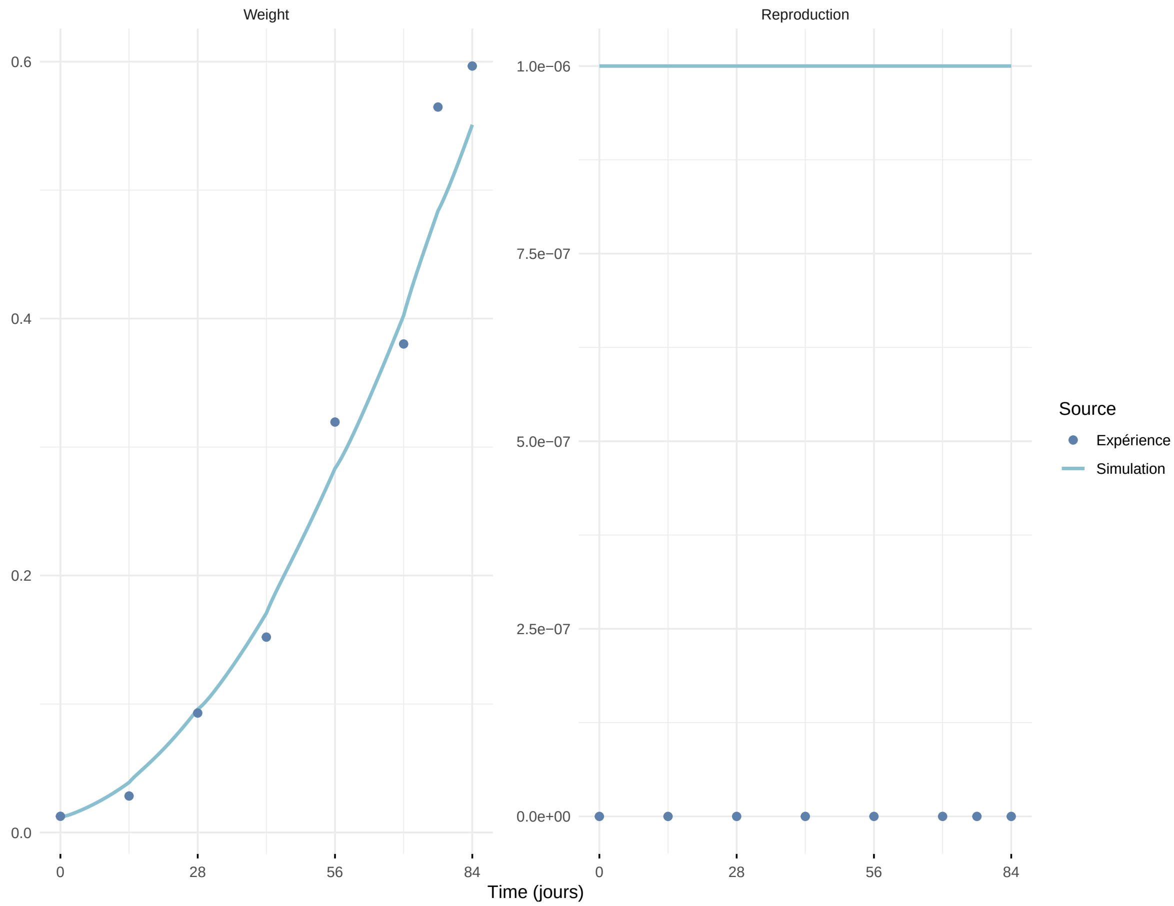
Expérience : Bart2019_XP1_1

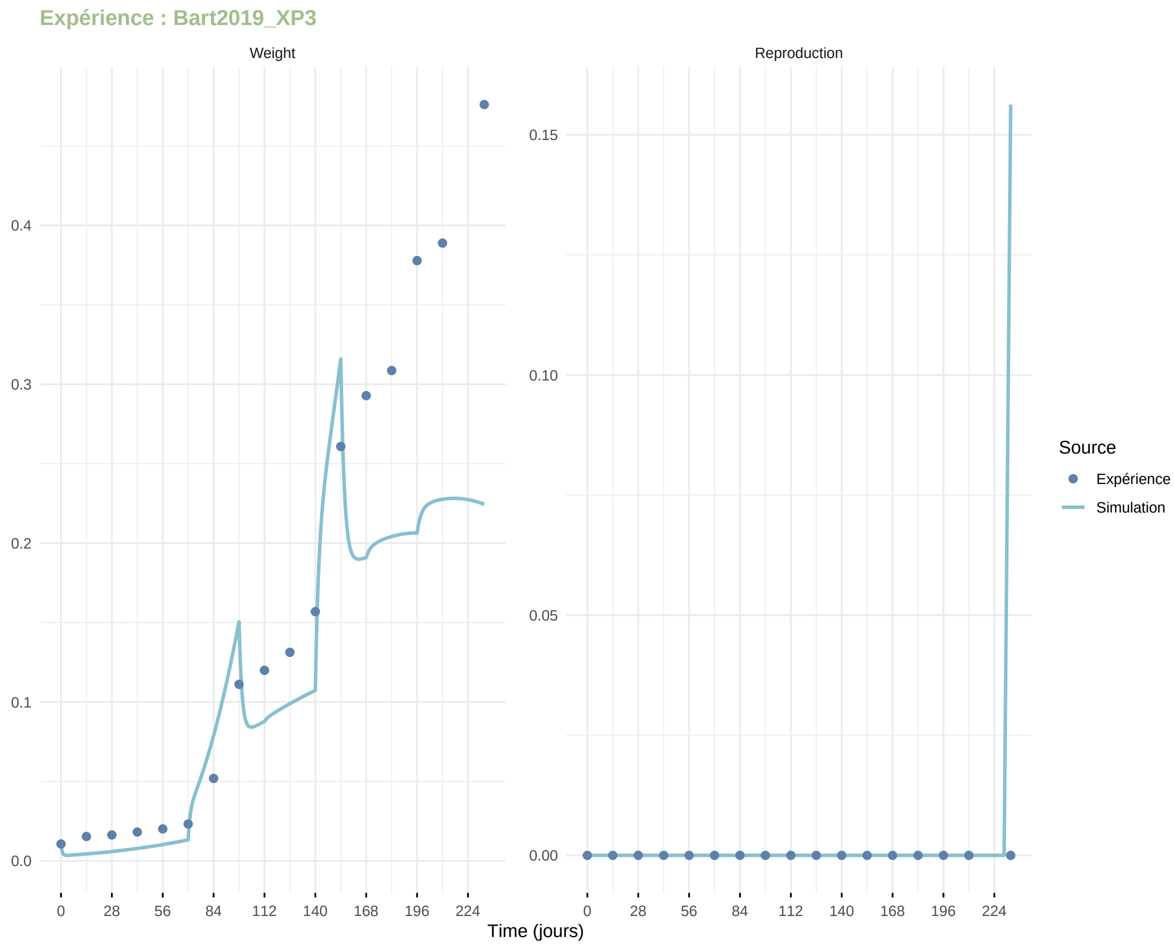


Expérience : Bart2019_XP1_2

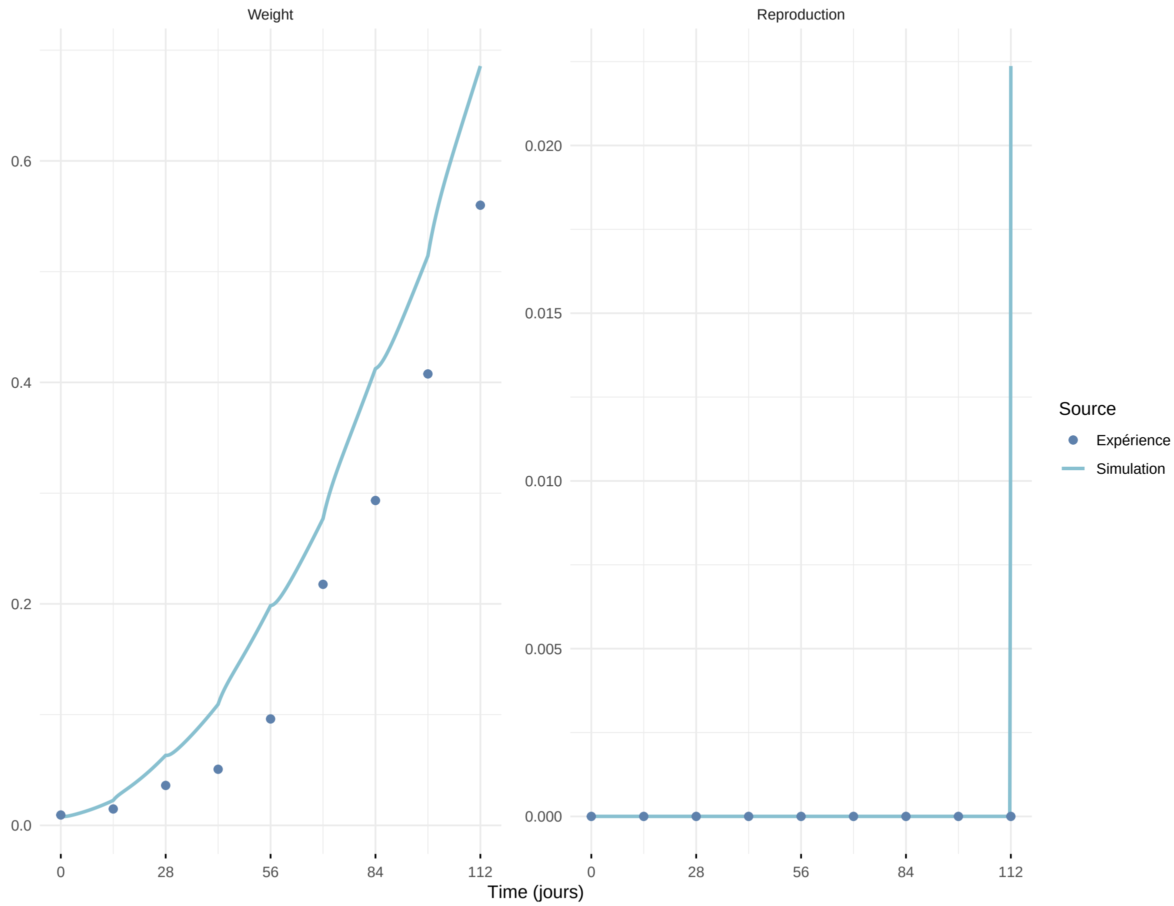


Expérience : Bart2019_XP2

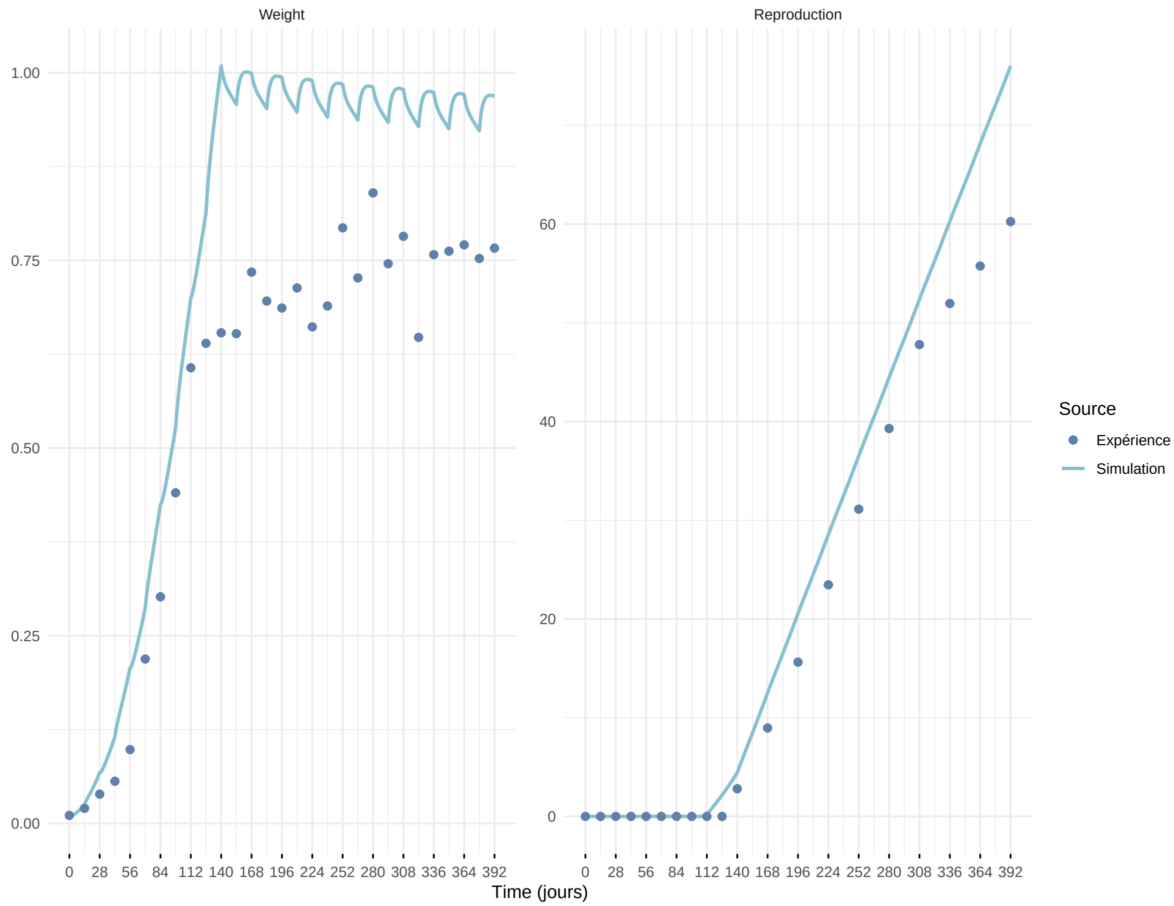



Expérience : Bart2019_XP3
Weight
Reproduction
0.0
0.1
0.2
0.3
0.4
0.00
0.05
0.10
0.15
0
28
56
84
112
140
168
196
224
Time (jours)
Source
Expérience
Simulation

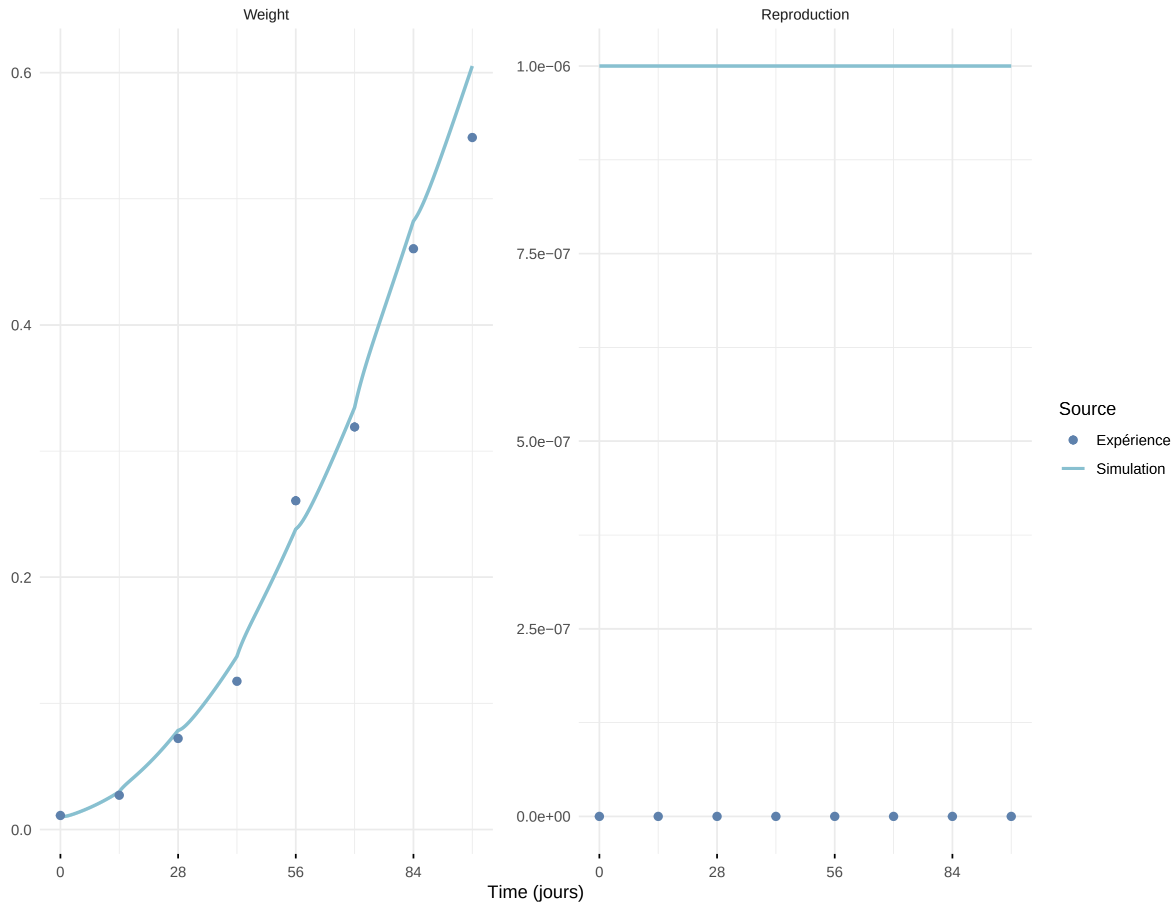
Expérience : Bart2019_XP4_1

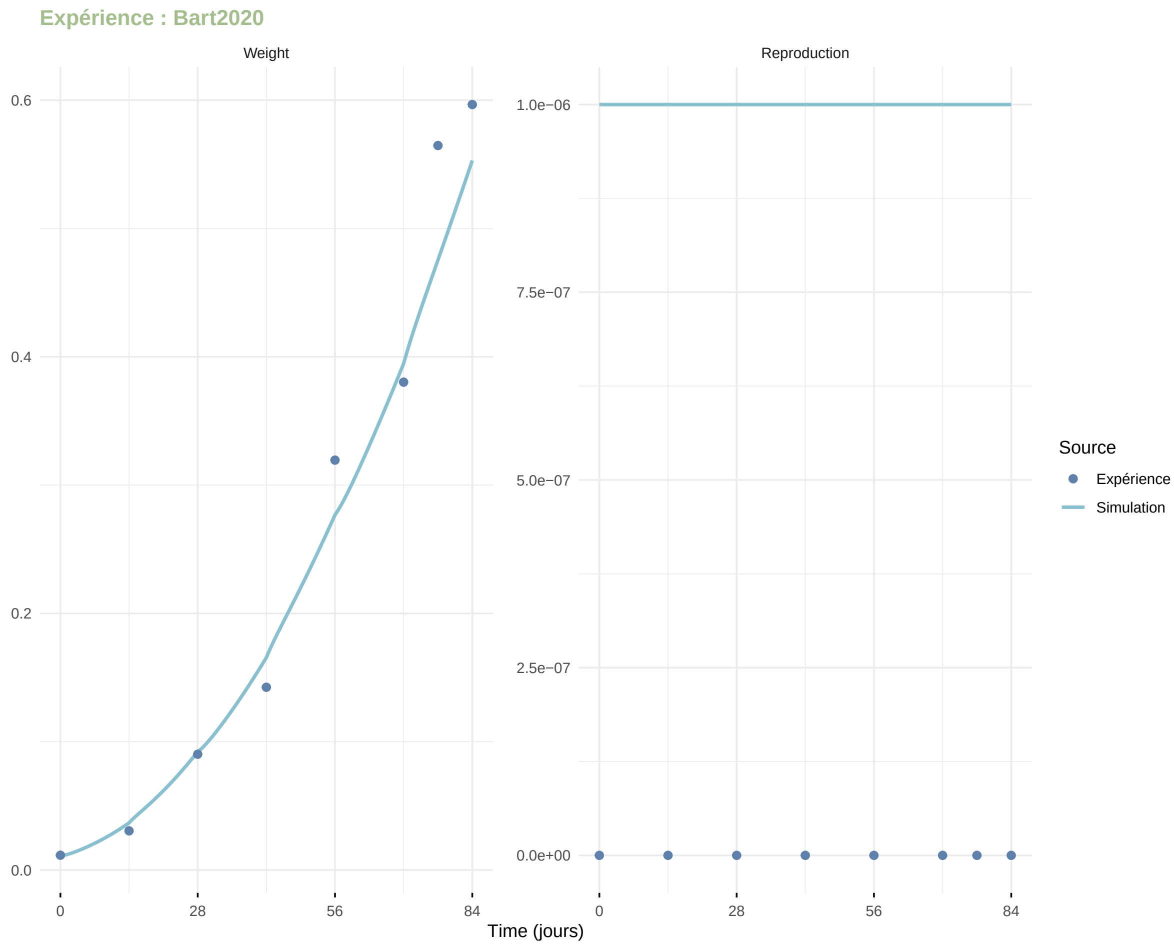


Expérience : Bart2019_XP4_2

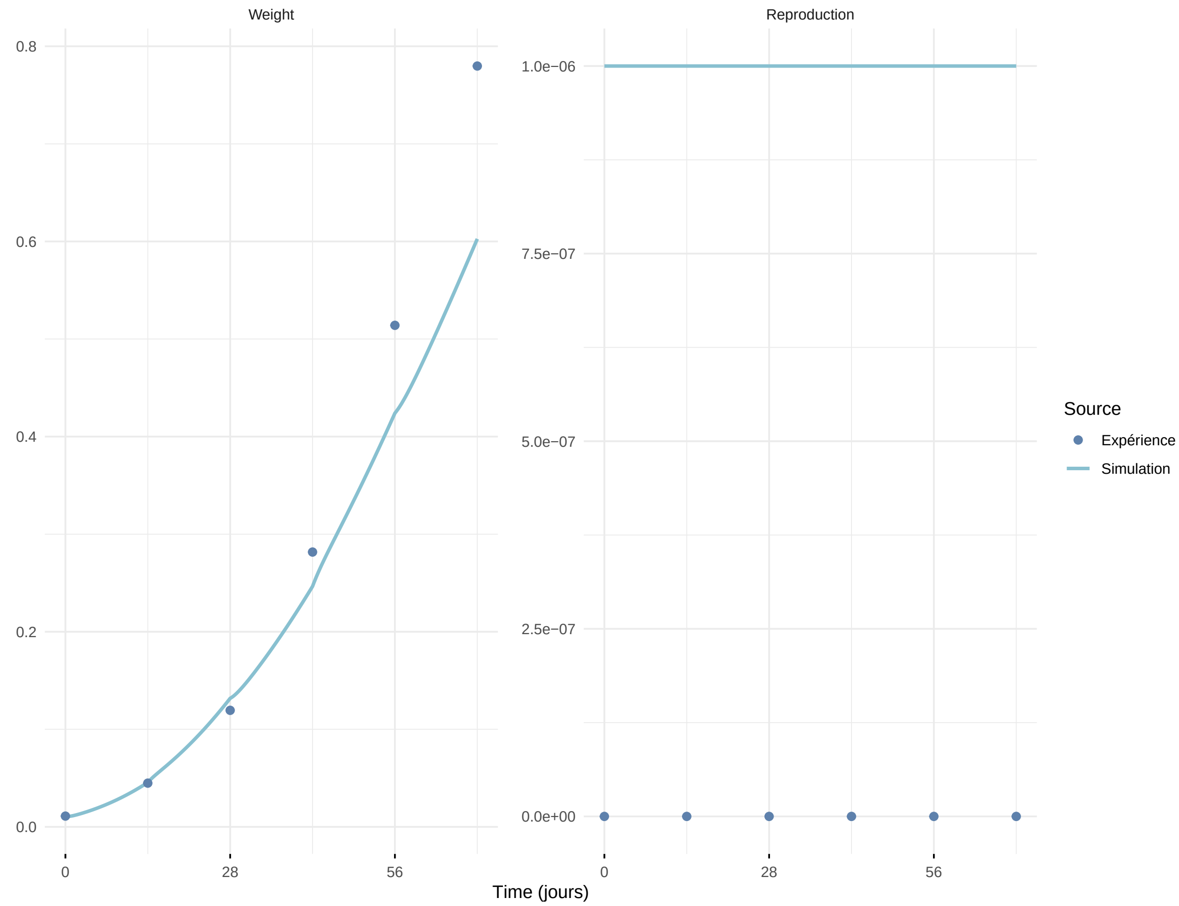


Expérience : Bart2019_XP5

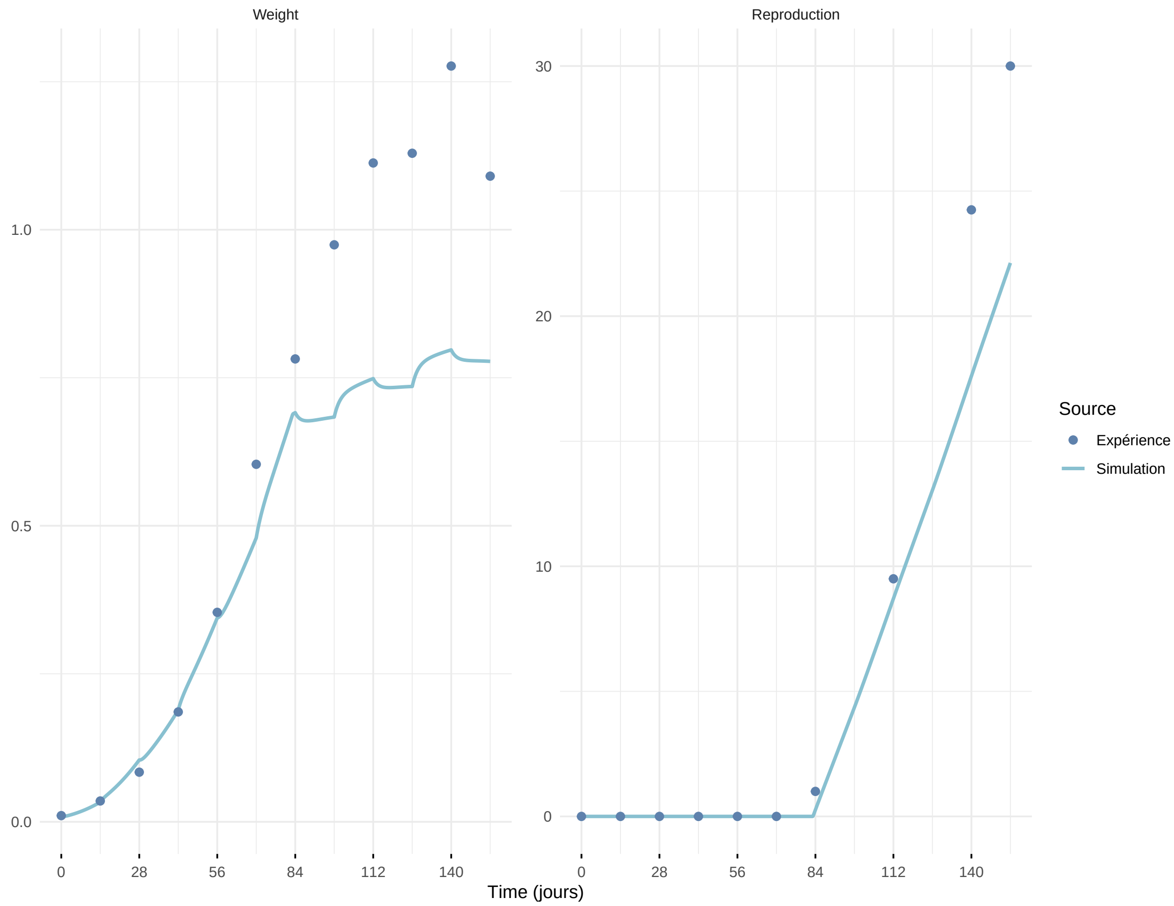



Expérience : Bart2020
Weight
Reproduction
0.6
0.4
0.2
0.0
1.0e−06
7.5e−07
5.0e−07
2.5e−07
0.0e+00
0
28
56
84
Time (jours)
Source
Expérience
Simulation

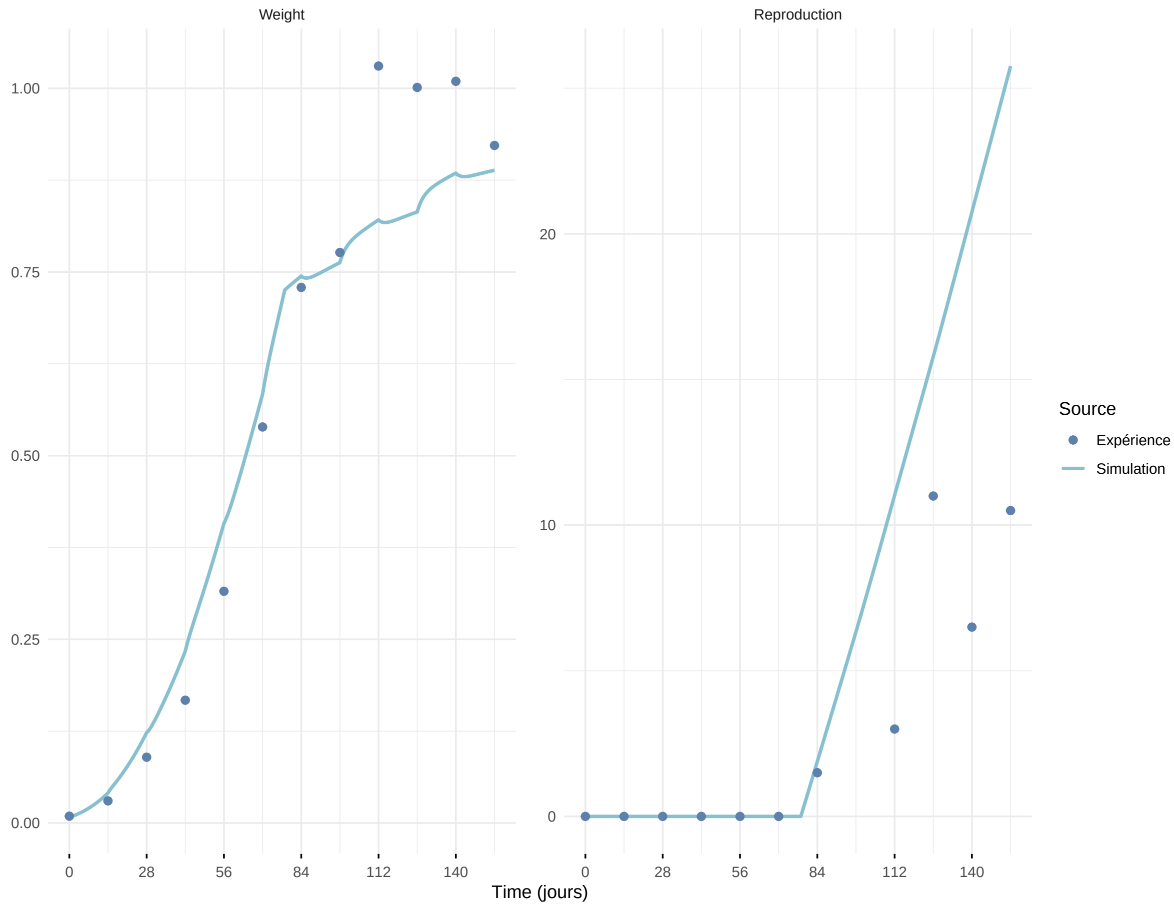
Expérience : Gollot2026_dens_D1



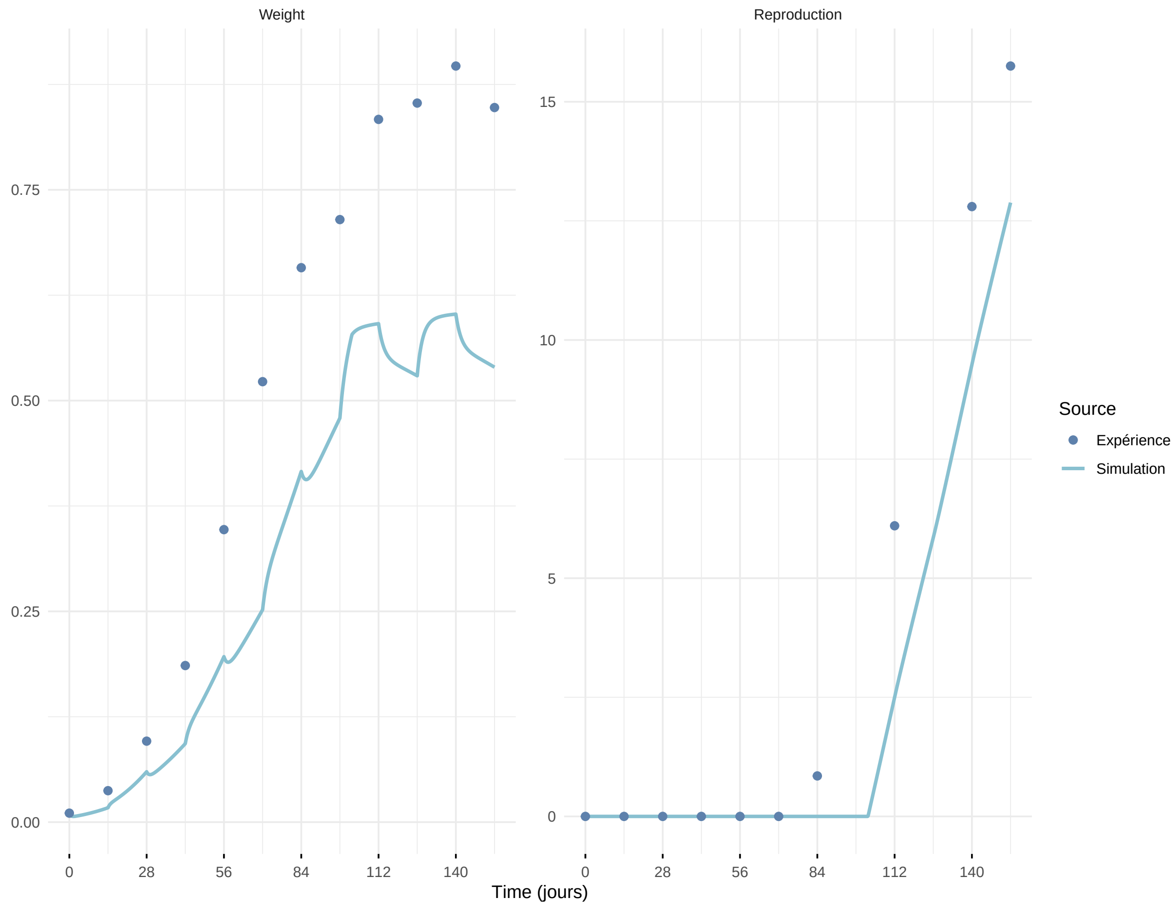
Expérience : Gollot2026_dens_D2

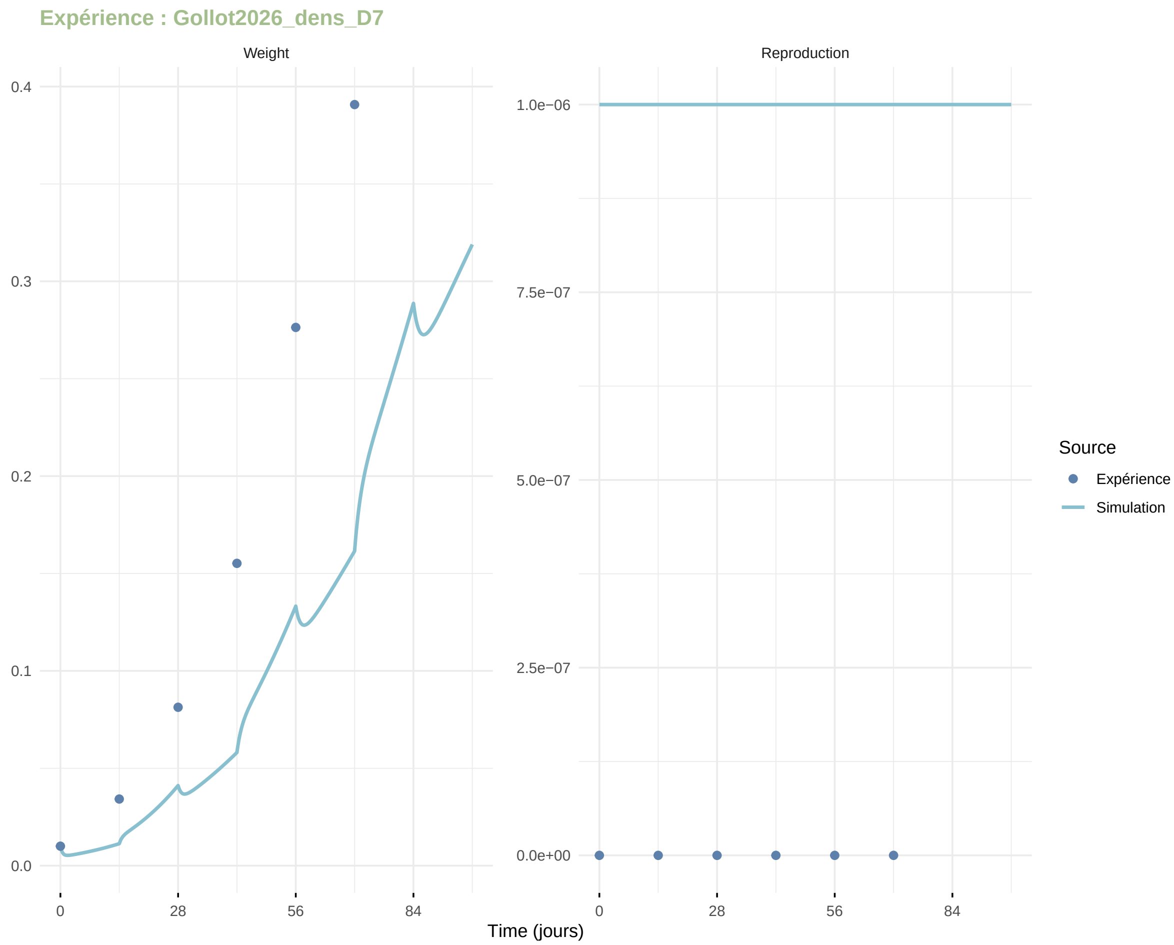


Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5




Expérience : Gollot2026_dens_D7
Weight
Reproduction
0.4
0.3
0.2
0.1
0.0
1.0e−06
7.5e−07
5.0e−07
2.5e−07
0.0e+00
0
28
56
84
Time (jours)
Source
Expérience
Simulation